(1) - Find the general solution of the given system of equations
- Describe the behavior as $t \to \infty$
- Draw a sketch w/ and w/o technology.

a) $X' = \begin{pmatrix} 5/4 & 3/4 \\ 3/4 & 5/4 \end{pmatrix} X$

$\det(A - \lambda I) = \det\begin{pmatrix} 5/4 - \lambda & 3/4 \\ 3/4 & 5/4 - \lambda \end{pmatrix}$

$= (5/4 - \lambda)(5/4 - \lambda) - 9/16$

$= 25/16 - 5/4\lambda - 5/4\lambda + \lambda^2 - 9/16$

$= \lambda^2 - 10/4\lambda - 16/16$

$= \lambda^2 - 5/2\lambda - 1 = 1/2(2\lambda^2 - 5\lambda - 2)$

$= 1/2(2\lambda - 1)(\lambda - 2)$

- set against 0

$(2\lambda - 1)(\lambda - 2)$, let $\lambda_1 = 1/2$ & $\lambda_2 = 2$

For $\lambda_1 = 1/2$ $\begin{pmatrix} 5/4 & 3/4 \\ 3/4 & 5/4 \end{pmatrix} \Rightarrow \begin{pmatrix} 3/4 & 3/4 \\ 3/4 & 3/4 \end{pmatrix}$

For $\lambda_2 = 2$ $\begin{pmatrix} 5/4 & 3/4 \\ 3/4 & 5/4 \end{pmatrix} \Rightarrow \begin{pmatrix} & 3/4 \\ 3/4 & \end{pmatrix}$

For $\lambda_1 = 1/2$ $\begin{pmatrix} 1/4 & 3/4 \\ 3/4 & 1/4 \end{pmatrix}\begin{pmatrix} V_{1,1} \\ V_{1,2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$V_{1,1} + V_{1,2} = 0 \Rightarrow V_{1,1} = -V_{1,2}$

let $V_{1,1} = 1$ then $V_{1,2} = -1$ & $V_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

For $\lambda_2 = 2$ $\begin{pmatrix} -3/4 & 3/4 \\ 3/4 & -3/4 \end{pmatrix}\begin{pmatrix} V_{2,1} \\ V_{2,2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$V_{2,1} = V_{2,2}$, let $V_{2,1} = 1$ then $V_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

- This gives a general solution of

$$X(t) = C_1 e^{2t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^{1/2 t}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- Describe the dominant behavior as $t \to \infty$

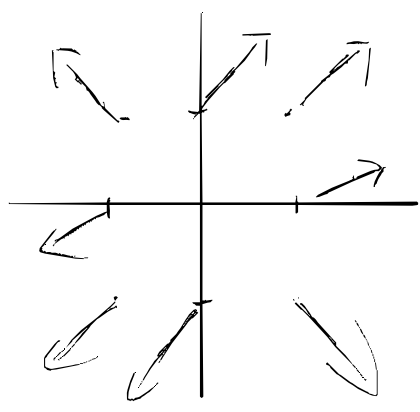
As $t \to \infty$, e^{2t} becomes dominant, so, we approach $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow y = x$

- unless on $y = -x$

- Sketch by hand

$X' = \begin{pmatrix} 5/4 & 3/4 \\ 3/4 & 5/4 \end{pmatrix} X \Rightarrow \begin{matrix} x' = 5/4 x + 3/4 y \\ y' = 3/4 x + 5/4 y \end{matrix}$

- we're only interested in relative rates, so, drop the /4's

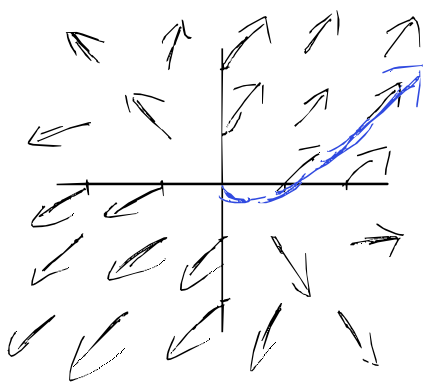


$x' \approx 5x + 3y$
$y' \approx 3x + 5y$

(a) $(0, 1)$, $x' = 3$, $y' = 5$
(b) $(1, 0)$, $x' = 5$, $y' = 3$
(c) $(1, 1)$, $x' = 8$ $y' = 8$
(a) $(-1, 1)$, $x' = -2$, $y' = 2$
(a) $(-1, 0)$, $x' = -5$, $y' = -3$
(b) $(-1, -1)$, $x' = -8$, $y' = -8$
(b) $(0, -1)$, $x' = -3$, $y' = -5$
(a) $(1, -1)$, $x' = 2$, $y' = -2$

- we can see the two eigen directions

- Sketch w/ tech.
- AM using a desmos calculator and reproducing



$F(x,y) = \left(\frac{5}{4}x + \frac{3}{4}y, \frac{3}{4}x + \frac{5}{4}y\right)$

(1) - Find the general solution of the given system of equations
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b) $X' = \begin{pmatrix} 4 & -3 \\ 8 & -6 \end{pmatrix} X$

$\det(A - \lambda I) = \det\begin{pmatrix} 4-\lambda & -3 \\ 8 & -6-\lambda \end{pmatrix}$

$= (4-\lambda)(-6-\lambda) + 24$

$= -24 - 4\lambda + 6\lambda + \lambda^2 + 24$

$= \lambda^2 + 2\lambda = (\lambda)(\lambda + 2)$

let $\lambda_1 = 0$ & $\lambda_2 = -2$

- For $\lambda_1 = 0$

$\begin{pmatrix} 4 & -3 \\ 8 & -6 \end{pmatrix}\begin{pmatrix} V_{1,1} \\ V_{1,2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ - scalar multiples

$4V_{1,1} - 3V_{1,2} = 0 \Rightarrow 4V_{1,1} = 3V_{1,2}$

let $V_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ From $V_{1,2} = \frac{4}{3}V_{1,1}$

- For $\lambda_1 = -2$

$\begin{pmatrix} 6 & -3 \\ 8 & -4 \end{pmatrix}\begin{pmatrix} V_{2,1} \\ V_{2,2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ - scalar multiples

$6V_{2,1} - 3V_{2,2} = 0 \Rightarrow 6V_{2,1} = 3V_{2,2}$

$\Rightarrow 2V_{2,1} = V_{2,2}$

let $V_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$X(t) = C_1\begin{pmatrix} 3 \\ 4 \end{pmatrix} + C_2 e^{-2t}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

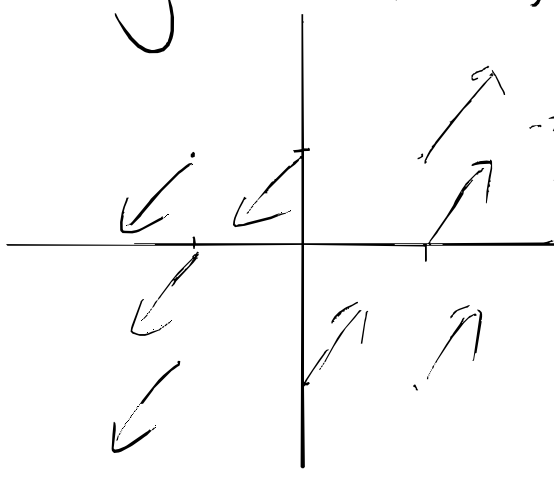
- Describe the behavior as $t \to \infty$

Even though $\lambda_1 = 0$ disappears, as $t \to \infty$, λ_1 is dominant over $\lambda_2 = -2$, as $t \to \infty$ then we'd expect $\lim_{t \to \infty} dy/dx \to \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

or $\lim_{t \to \infty} dy/dx \to \begin{pmatrix} -3 \\ -4 \end{pmatrix}$,

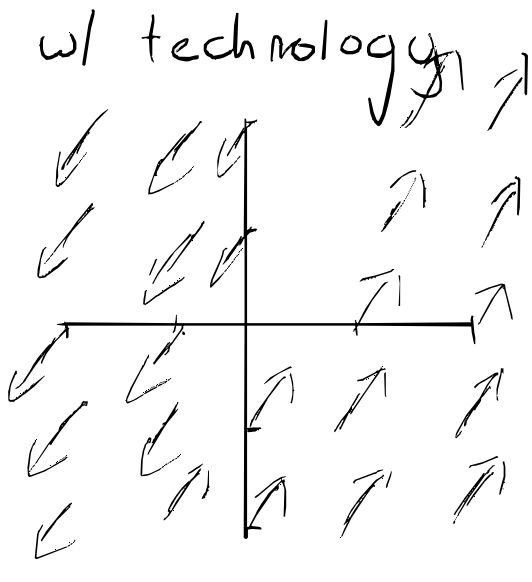
depending on initial conditions

- Plot by hand. $\begin{pmatrix} 4 & -3 \\ 8 & -6 \end{pmatrix} \Rightarrow \begin{matrix} x' = 4x - 3y \\ y' = 8x - 6y \end{matrix}$



$\odot$ $(0,0)$, $x' = 0$, $y' = 0$
-2 $\odot$ $(0,1)$, $x' = -3$, $y' = -6$
2 $\odot$ $(1,0)$, $x' = 4$, $y' = 8$
-2 $\odot$ $(-1,0)$, $x' = -4$, $y' = -8$
2 $\odot$ $(0,-1)$, $x' = 3$, $y' = 6$
2 $\odot$ $(1,1)$, $x' = 1$, $y' = 2$
2 $\odot$ $(1,-1)$, $x' = 7$, $y' = 14$
-2 $\odot$ $(-1,1)$, $x' = -7$, $y' = -14$
-2 $\odot$ $(-1,-1)$, $x' = 1$, $y' = -2$

- Plot w/ technology



- again, simple desmos calculator

(1) - Find the general solution of the given system of equations
- Describe the behavior as $t \to \infty$
- Draw a sketch w/ and w/o technology.

(c) $x' = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix} x$

$\det \begin{pmatrix} 3-\lambda & -2 \\ 4 & -1-\lambda \end{pmatrix}$

$= (3-\lambda)(-1-\lambda) + 8 = \lambda^2 - 2\lambda - 3 + 8 = 0$

$= \lambda^2 - 2\lambda + 5$

$\lambda = \frac{2 \pm \sqrt{4-20}}{2} = 1 \pm \sqrt{-16}/2$

$= 1 \pm 4i/2 = 1 \pm 1 \pm 2i$, let $\lambda_1 = 1+2i$

$\begin{pmatrix} 3-1-2i & -2 \\ 4 & -1-1-2i \end{pmatrix} = \begin{pmatrix} 2-2i & -2 \\ 4 & -2-2i \end{pmatrix}$

$\begin{pmatrix} 2-2i & -2 \\ 4 & -2-2i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$(2-2i)(v_1) - 2v_2 = 2v_1 - 2iv_1 - 2v_2 = 0$

$2v_1 - 2iv_1 = 2v_2 \Rightarrow v_1 - iv_1 = v_2$

$\Rightarrow (1-i)v_1 = v_2$, let $v_1 = 1$, then

$v_2 = 1-i$, so $v = \begin{pmatrix} 1 \\ 1-i \end{pmatrix}$

- By Euler: $e^{(a+bi)t} = e^{at}(\cos(bt) + i\sin(bt))$
- So for $\lambda_1 = 1+2t$ we get

$e^{(1+2i)t} = e^{1t}(\cos(2t) + i\sin(2t))$

- From above

$ve^{(1+2i)t} = e^t \begin{pmatrix} 1 \\ 1-i \end{pmatrix} (\cos(2t) + i\sin(2t))$

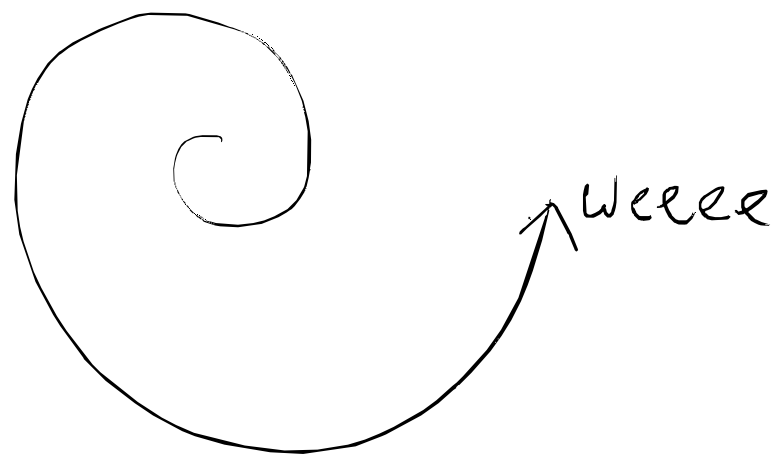
- Multiply it out

$= e^t \begin{pmatrix} \cos(2t) + i\sin(2t) \\ \cos(2t) + i\sin(2t) - i\cos(2t) + \sin(2t) \end{pmatrix}$

- Splitting into real & imaginary

$x(t) = c_1 e^t \begin{pmatrix} \cos(2t) \\ \cos(2t) + \sin(2t) \end{pmatrix} + c_2 e^t \begin{pmatrix} \sin(2t) \\ \sin(2t) - \cos(2t) \end{pmatrix}$

- Behavior as $t \to \infty$
  - Well, it's complex, so a spiral, as $Re(\lambda) > 1$ we're an unstable



- More plotting by hand...
- Lazy programmer Flex

seq(seq(i, j, -2, 2), i, -2, 2) -> x

seq(seq(j, j, -2, 2), i, -2, 2) -> y

3x - 2y -> dx

4x - y -> dy

10 • (dx ./ (abs(dx))) + (dy ./ (abs(dy)))

Gives a 5x5 w/ 11 = ↗

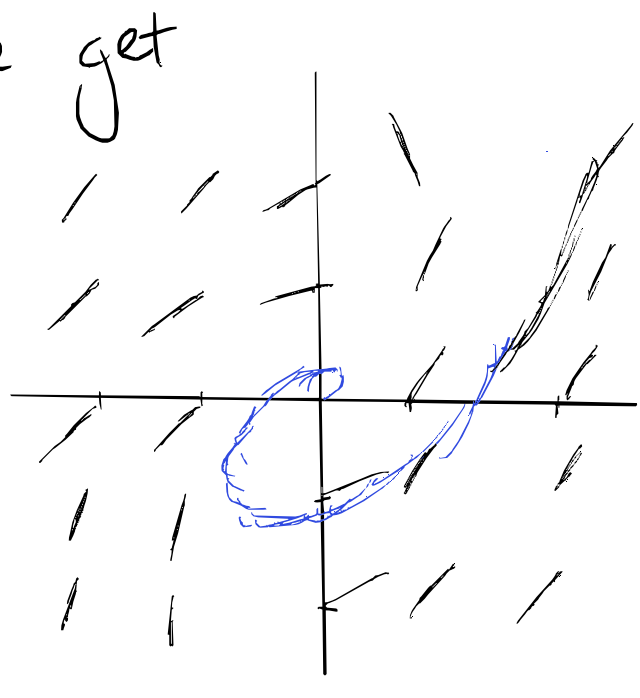
$\begin{pmatrix} \swarrow & \swarrow & \swarrow & \swarrow & \swarrow \\ \searrow & \swarrow & \swarrow & \swarrow & \swarrow \\ \nearrow & \nearrow & & \swarrow & \swarrow \\ \nearrow & \nearrow & \nearrow & \nearrow & \nwarrow \\ \nearrow & \nearrow & \nearrow & \nearrow & \nearrow \end{pmatrix}$

9 = ↘

-9 = ↖

-11 = ↙

- I'm counting that as by hand, it's definetly more fun
- If we throw it in to desmos we get



(1) - Find the general solution of the given system of equations
- Describe the behavior as $t \to \infty$
- Draw a sketch w/ and w/o technology.

a) $X' = \begin{pmatrix} 2 & -5/2 \\ 9/5 & -1 \end{pmatrix}$

$$\det(A - \lambda I_2) = \det\begin{pmatrix} 2-\lambda & -5/2 \\ 9/5 & -1-\lambda \end{pmatrix}$$

$$= (2-\lambda)(-1-\lambda) + 9/2$$

$$= \lambda^2 - \lambda + 5/2 = 0 = 2\lambda^2 - 2\lambda + 5$$

$$\frac{2 \pm \sqrt{4-40}}{4} = \frac{2 \pm \sqrt{-36}}{4} = 1/2 \pm 3/2\, i$$

lets let $\lambda_1 = 1/2 + 3/2\, i$

$$\begin{pmatrix} 2 - 1/2 - 3/2\, i & -5/2 \\ 9/5 & -1 - 1/2 - 3/2\, i \end{pmatrix} = \begin{pmatrix} 3/2 - 3/2\, i & -5/2 \\ 9/5 & -3/2 - 3/2\, i \end{pmatrix}$$

$$\begin{pmatrix} 3/2 - 3/2\, i & -5/2 \\ 9/5 & -3/2 - 3/2\, i \end{pmatrix}\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$V_1(3/2 - 3/2\, i) - 5/2\, V_2 = 0$$

$$3/2\, V_1 - 3/2\, V_1 i = 5/2\, V_2$$

$$V_1(3/2 - 3/2\, i) = 5/2\, V_2$$

$$V_1\left(\frac{\cancel{2}}{5} \cdot \frac{3}{\cancel{2}} - \frac{\cancel{2}}{5} \cdot \frac{3}{\cancel{2}} i\right) = V_2 = (3/5 - 3/5\, i) V_1$$

let $V_1 = 1$, then $V_2 = 3/5 - 3/5\, i$

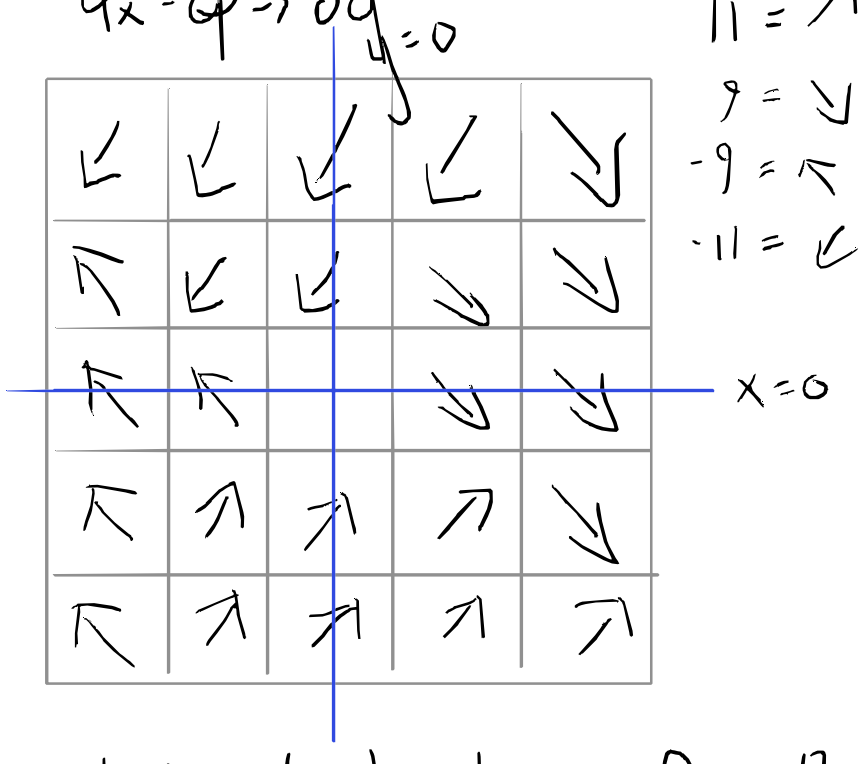
$$V = \begin{pmatrix} 5/2 \\ 3/2 - 3/2\, i \end{pmatrix}$$

$$X(t) = c_1 e^{1/2 t}\begin{pmatrix} 5/2 \cos(3/2 t) \\ 3/2 \cos(3/2 t) + 3/2 \sin(3/2 t) \end{pmatrix} + c_2 e^{1/2 t}\begin{pmatrix} 5/2 \sin(3/2 t) \\ 5/2 \sin(3/2 t) - 3/2 \sin(3/2 t) \end{pmatrix}$$

- Describe the behavior as $t \to \infty$
  - We know we have a spiral as $a > 0$ For e^{at}, we know it to be unstable
- Phase portrait by hand

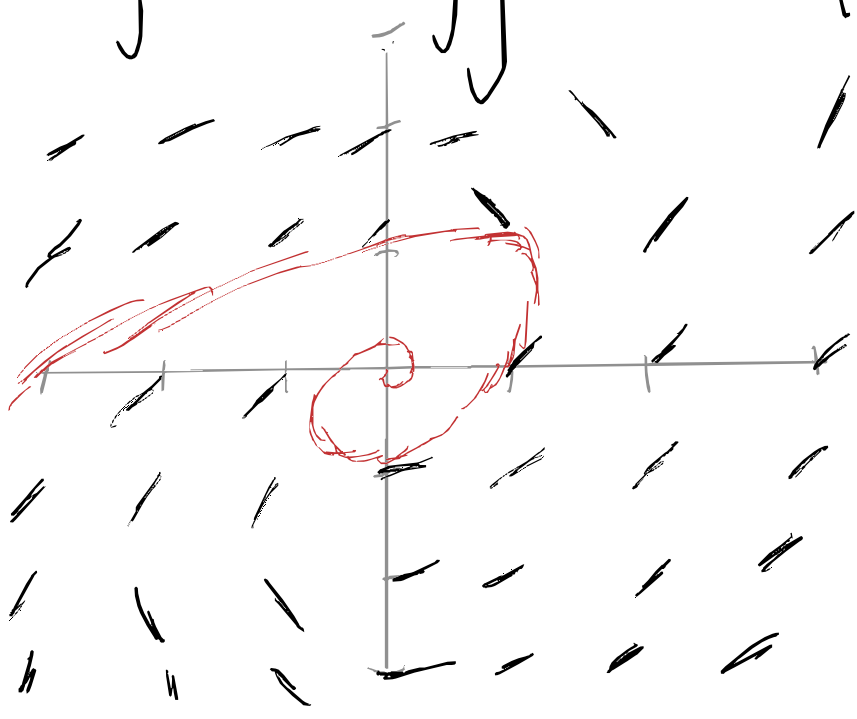
seq(seq(i, j, -2, 2), i, -2, 2) → x
seq(seq(j, j, -2, 2), i, -2, 2) → y

3x - 2y → dx
4x - y → dy



11 = ↗
9 = ↘
-9 = ↖
-11 = ↙

- Using technology for the plot



(2) Find the general solution

$$X'(t) = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix} X(t)$$

$$\det \begin{pmatrix} 1-\lambda & 1 & 2 \\ 1 & 2-\lambda & 1 \\ 2 & 1 & 1-\lambda \end{pmatrix}$$

$$= (1-\lambda)\left[(2-\lambda)(1-\lambda)-1\right]$$
$$-\left[(1-\lambda)-2\right] + (2)\left[1-(2)(2-\lambda)\right]$$
$$= (1-\lambda)\left[2-2\lambda-\lambda+\lambda^2-1\right]$$
$$-(-\lambda-1) + (2)\left[1-4-2\lambda\right]$$
$$= (1-\lambda)(\lambda^2-3\lambda+1)$$
$$+(\lambda+1) + (2)(2\lambda-3)$$
$$= (\lambda^2-3\lambda+1-\lambda^3+3\lambda^2-\lambda)$$
$$+(\lambda+1)+4\lambda-6$$
$$= (-\cancel{\lambda^3}+4\lambda^2-4\lambda+1)+\lambda+1+4\lambda-6$$
$$= \qquad\qquad = 0$$

$$= (-\lambda^3+4\lambda^2)+(\lambda-4) = 0$$
$$= \lambda^2(-\lambda+4)+(\lambda-4)$$
$$= -\lambda^2(\lambda-4)+(\lambda-4)$$

$$-\lambda^3+4\lambda^2+\lambda-4$$

	λ^3	λ^2	λ^1	λ^0
4	-1	4	1	-4
		-4	0	4
	-1	0	1	0

$$-(\lambda-4)(\lambda-1)(\lambda+1) = 0$$

let $\lambda_1 = 4, \lambda_2 = 1, \lambda_3 = -1$

$$\lambda_1 = 4: \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} -3 & 1 & 2 \\ 1 & -2 & 1 \\ 2 & 1 & -3 \end{pmatrix}$$

$$\lambda_2 = 1: \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix}$$

$$\lambda_3 = -1 \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 2 \end{pmatrix}$$

$$\lambda_1 = 4: x_1 = x_2 \;\&\; x_2 = x_3 \Rightarrow V_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 1: x_1 = x_2 \;\&\; x_2 = -2x_3 \Rightarrow V_2 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

$$\lambda_3 = -1: x_1 = -x_3 \;\&\; x_2 = 0 \Rightarrow V_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$X(t) = C_1 e^{4t}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + C_2 e^{t}\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + C_3 e^{-t}\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

(3) Consider $ay''+by'+cy=0$ where a,b,c are constants w/ $a \neq 0$, then the general solution depends on the roots of the characteristic $ar^2+br+c=0$

a) Transform $ay''+by'+cy=0$ into a system of first order equations by letting $x_1=y$, $x_2=y'$. Then find the system of equations $x'=Ax$ satisfied by $x=\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$$ay''+by'+cy=0$$

$$ay''=-by'-cy$$

$$y''=-\frac{b}{a}y'-\frac{c}{a}y$$

substitute $\Rightarrow x_2'=-\frac{b}{a}x_2-\frac{c}{a}x_1$

$$=-c/a\,x_1-b/a\,x_2$$

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix}=\begin{pmatrix} 0 & 1 \\ -c/a & -b/a \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$A=\begin{pmatrix} 0 & 1 \\ -c/a & -b/a \end{pmatrix}$$

b) Find the equation that'd give the eigenvalues of the coefficient matrix

$$\det\left[\begin{pmatrix} 0 & 1 \\ -c/a & -b/a \end{pmatrix}-\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right]=0$$

$$=\det\begin{pmatrix} -\lambda & 1 \\ -c/a & -b/a-\lambda \end{pmatrix}=0$$

$$=(-\lambda)(-b/a-\lambda)+c/a=0$$

$$=\lambda b/a+\lambda^2+c/a=0$$

$$=\lambda^2+\lambda b/a+c/a=0 \Rightarrow a\lambda^2+b\lambda+c=0$$

(4) Consider the system

$$x'(t) = \begin{pmatrix} -1 & -1 \\ -\alpha & -1 \end{pmatrix} x$$

a) Solve the system for $\alpha = 1/2$

$$\det \begin{pmatrix} -1-\lambda & -1 \\ -\alpha & -1-\lambda \end{pmatrix} = 0$$

$$= (1+\lambda)(1+\lambda) - \alpha$$

$$= \lambda^2 + 2\lambda + 1 - \alpha$$

$$x = \frac{-8 \pm \sqrt{4 - 4(1-\alpha)}}{8}$$

$$= -1 \pm \sqrt{1 - 1 + \alpha}$$

$$= -1 \pm \sqrt{\alpha}$$

let $\lambda_1 = -1 + \sqrt{1/2}$, $\lambda_2 = -1 - \sqrt{1/2}$

$\lambda_1 < 0$ & $\lambda_2 < 0$

We'd therefore have a stable sink, though we'd converge more to one axis than the other as $\lambda_1 \neq \lambda_2$

b) Repeat for $\alpha = 2$

$$= -1 \pm \sqrt{\alpha}$$

let $\lambda_1 = -1 + \sqrt{2}$ & $\lambda_2 = -1 - \sqrt{2}$

$\lambda_1 > 0$ & $\lambda_2 < 0$

We'd expect to see a saddle here, one is growing while the other shrinks

c) Find where the transition happens between saddle and stable sink.

We need the value at which $-1 \pm \sqrt{\alpha}$ transitions, this is easily spotted, $\alpha = 1$, here let $\lambda_1 = -2$, $\lambda_2 = 0$, we'd expect to see degeneracy at the exact value.